Review Report for *Differentiable Discrete Event Simulation Applications in Queuing Network Control*

This paper introduces a novel framework for policy optimization in queueing network control using differentiable discrete event simulation. The main contribution lies in the careful design of smoothing techniques to compute pathwise gradients. The paper is well-written, and I enjoyed reading it. I appreciate the authors' effort in combining modern machine learning methods with domain knowledge to address important operations research problems. However, I still have concerns regarding the empirical evaluation and the positioning of this paper within the existing literature.

1. The gradient estimation methods proposed in this paper are not new. The fractional allocation technique is standard and the 'straight-through' trick exists in the literature [Bengio et al., 2013]. Therefore, the authors should better position this paper in the literature.

2. Since the main contributions lie in the application to queueing systems, I believe the empirical evaluation should be conducted more thoroughly and described in greater detail. I also hope the authors will consider open-sourcing their code.

   - I am surprised that the parameters (service rates, arrival rates, cost, etc) of the numerical examples (reentrants, and criss-cross networks) are not specificed. Moreover, several important implementation details are missing or unclear. For example, the discount factor γ used in the REINFORCE method is not mentioned in Section 4.1. Additionally, it is unclear how the baseline term $V_{\pi_\theta}(x_k)$ is constructed in the REINFORCE implementation.
   - This paper focuses on work-conserving policies and argues that work conservation serves as a powerful inductive bias. However, in the queueing literature, non-work-conserving policies can also be optimal under certain conditions. For instance, such policies are shown to be optimal in specific parameter

regimes of criss-cross networks; see Cases IIB and IID in Martins et al. [1996] and Budhiraja et al. [2017]. Can the proposed method be generalized to handle these settings as well?

Remark: I agree with the authors that "standard model-free RL algorithms are based on the 'tabula rasa' principle, which aims to search over a general and unstructured policy class to find an optimal policy." In this context, I believe the ability to leverage the known recursive form $s_{k+1} = f(s_k, u_k, \xi_{k+1})$ or the Aux Update is key to the success of the proposed method. Given this structure, I believe the method has the potential to discover non-work-conserving optimal policies as well.

- The theoretical analysis suggests that the heavy-traffic regime (i.e., $\rho \to 1$) represents the most challenging case. Arguably, this regime is also the most interesting from an empirical perspective. I would be interested in seeing how the proposed method performs as ρ increases. Additionally, several recent works have explored diffusion control problems in the heavy-traffic regime using neural networks [Han et al., 2018, Ata et al., 2024, Ata and Kaşıkaralar, 2023]. It would be valuable to discuss and, if possible, compare the proposed approach with these methods.

3. A minor writing suggestion: I recommend moving the criss-cross example earlier in the text to improve the flow. Currently, Figures 1 and 2 are based on the criss-cross example, but the example itself is introduced only later. Reordering this would help readers better understand the figures when they first appear.

References

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